

Total No. of Questions : 5]

SEAT No. :

PD-2847

[Total No. of Pages : 3

[6432]-2002

M.C.A.

MANAGEMENT

**STQ - 553 MJ : Software Testing and Quality Assurance
(2024 Pattern) (Semester - II)**

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw daigram wherever necessary.*

- Q1)** a) Describe software quality assurance and outline its key activities. [5]
b) Explain the metrics used to assess and measure reliability. [5]

OR

- c) Explain the term quality in the context of software engineering and describe product revision factors. [5]
d) Describe the significance of ISO 9000 standards in SQA. [5]

- Q2)** a) Outline the steps in software testing life cycle, highlighting the sequence of testing phases. [5]

- b) Differentiate between fault, defect and failure and describe the common causes of software failure. [5]

OR

- c) Illustrate the concept of performance testing and detail any three specific performance test categories. [5]

- d) Discuss the Integration test approaches. [5]

- Q3)** a) In an examination, a candidate must score at least 24 marks to pass, with the maximum possible score being 40 marks.

Using the Equivalence partitioning technique:

- i) identify the equivalence partitions for the marks field.
- ii) determine the minimum number of test cases required to effectively test the field.
- iii) Specify the test cases, including representative values and their expected outcomes. [5]

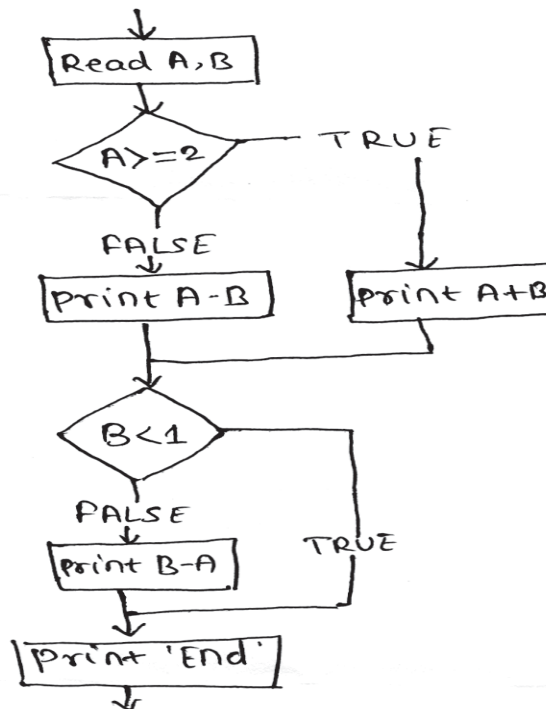
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- b) Differentiate between static and dynamic testing. [5]

OR

- c) Given the following flow chart diagram:

What is the minimum number of test cases required for 100% statement coverage and 100% decision coverage? Justify the answer. [5]



- d) Discuss inspection process in detail. [5]

Q4) a) Cloud kitchen aggregator app, smoke & spice lists and promotes cloud kitchens, making them accessible to customers for online food ordering. It enables cloud kitchens to manage their menus, orders, and promotions while integrating with third party delivery partners and payment gateways. The app challenges are handling high-order volumes, real time order updates, seamless third-party integrations, and secure transactions.

A Write a test plan for the following sections of the IEEE 829 test plan template for the above application. [10]

- i) scope of testing
- ii) objectives
- iii) risks
- iv) strategy
- v) approach

OR

- b) Write suitable test case for the above application. [10]

- Q5)** a) Define CAST and explain its benefits in software testing. [5]
b) Describe the steps in introducing a testing tool into an organization. [5]
OR
c) Explain the purpose of postman in API testing. [5]
d) Discuss defect life cycle. [5]

